

To: Dr. Hannah Lee (Primary Care)
From: Hematology/Oncology Service
Date: 04.09.2025
Patient: Carol Scott (02.12.1977), MRN AML030
Diagnosis: Acute Myeloid Leukemia, ELN 2022 Favorable, t(8;21) (q22;q22.1) producing RUNX1-RUNX1T1 fusion
Initial therapy: Standard "7+3" induction 12.12.2022 followed by high-dose cytarabine consolidation

Presentation & Diagnostics

Ms. Scott presented in December 2022 with gingival swelling, fatigue, and easy bruising. Initial labs: WBC $14.3 \times 10^9/L$ (blasts 35%), Hgb 7.9 g/dL, Plt $52 \times 10^9/L$, LDH 560 U/L. Coagulation normal; D-dimer mildly elevated. CT chest/abdomen without organomegaly.

Bone marrow 05.12.2022: 60–70% blasts; flow positive for CD34, CD117, HLA-DR, CD13/33; aberrant CD19 coexpression, consistent with **core-binding factor AML** phenotype.

Cytogenetics: t(8;21)(q22;q22.1); otherwise diploid.

Molecular: FLT3-ITD/TKD negative; NPM1 negative; KIT D816 not detected; CEBPA wild type.

Echocardiogram: LVEF 63% prior to anthracycline.

Treatment & Hospital Course

- **Induction (12.12.2022):** Cytarabine continuous infusion $\times 7$ days + daunorubicin $\times 3$ days.
 - **Complications:** Febrile neutropenia (cultures negative) treated empirically; cytarabine conjunctivitis prevented with intensive steroid eye drops; grade 2 mucositis and transient transaminitis (ALT peak 86 U/L).
 - **Day 28 marrow (10.01.2023):** Complete remission (CR), blasts $< 2\%$, MRD negative by multiparameter flow ($< 0.1\%$). **RUNX1-RUNX1T1 RT-qPCR** low-level but **> 3 -log reduction** from baseline.
- **Consolidation:** High-dose cytarabine (HiDAC) at $3 \text{ g/m}^2 \text{ q12h}$ on days 1,3,5 $\times 4$ cycles (Feb–June 2023).
 - Toxicities included expected cytopenias; one admission for neutropenic fever (April 2023, rhinovirus) without bacterial growth; no cerebellar toxicity (daily neuro checks normal); conjunctivitis prophylaxis continued.
 - **PCR tracking:** After cycle 2 (March 2023) a **> 4 -log reduction**; after cycle 4 (June 2023) **undetectable** by our assay.
- **Central line:** Right IJ tunneled catheter placed Dec 2022, electively removed July 2023.

Current Status & Surveillance

Ms. Scott has remained clinically well since completing consolidation. She returned to her work as a high-school art teacher and reports normal activity levels.

Latest Clinic Visit: 03.09.2025

Vitals: BP 116/72, HR 70, afebrile, BMI 24.8; ECOG 0.

Exam: No lymphadenopathy; lungs clear; abdomen benign; no petechiae.

Labs (03.09.2025): WBC $5.2 \times 10^9/L$ (ANC 2.5), Hgb 12.9 g/dL, Plt $254 \times 10^9/L$, MCV 91; creatinine 0.82 mg/dL; AST/ALT 19/17; ferritin 210 ng/mL.

Peripheral smear: Normal morphology.

Bone marrow & MRD: We have not required routine marrows since mid-2024 given persistently negative PCR and stable counts; the most recent **RUNX1-RUNX1T1 RT-qPCR (peripheral blood, July 2025)** remained **undetectable**.

Imaging: Not indicated; prior incidental hepatic steatosis on ultrasound improved with weight loss.

Comorbidities & Medications

- *Mild exercise-induced asthma (albuterol PRN).*
- *Hashimoto's hypothyroidism (**levothyroxine 50 µg qAM**).*
- *Vitamin D insufficiency (cholecalciferol 1000 IU daily).*
- *Copper IUD in situ; menstrual cycles light and regular.*

No anticoagulation. She avoids NSAIDs for occasional headaches, using acetaminophen instead.

Late Effects & Survivorship

- **Cardiac:** Anthracycline exposure tracked; **echo 08.2024 LVEF 61%**, asymptomatic; next surveillance echo planned in 2026.
- **Neuro/ocular:** No cerebellar signs; vision normal.
- **Fertility/menstrual:** Counseling provided; no amenorrhea.
- **Bone health:** Weight-bearing exercise; baseline DEXA normal.
- **Vaccinations:** Up to date (influenza annually; COVID boosters; pneumococcal per schedule).
- **Psychosocial:** No depression/anxiety; resumed teaching and painting; occasional fatigue at term start mitigated with sleep hygiene.

Plan Going Forward

1. **Surveillance:** Clinic and CBC every 3 months through end of 2025, then extend to every 4–6 months if stable. Continue **RUNX1-RUNX1T1 qPCR** from peripheral blood semiannually for two more years; if rising transcript detected, we will re-stage with marrow promptly.

2. **No maintenance chemotherapy** planned given sustained MRD negativity and favorable risk profile.
3. **When to call:** $T \geq 38.0^{\circ}\text{C}$, unexpected bruising/bleeding, new lymphadenopathy, exertional dyspnea, or recurrent infections.
4. **General health:** Continue routine cancer screening; dental care with hemostasis precautions if thrombocytopenic in the future (not expected).
5. **If relapse were ever suspected:** We would re-induce per CBF-AML protocols and consider clinical trials; transplant is not indicated in current remission but would be reconsidered based on disease biology.

Ms. Scott understands and agrees with the plan. Please contact us if any issues arise.

*Warm regards,
[Name], MD
Consultant Hematologist*